Lecture 25: Lines and Planes in Space

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

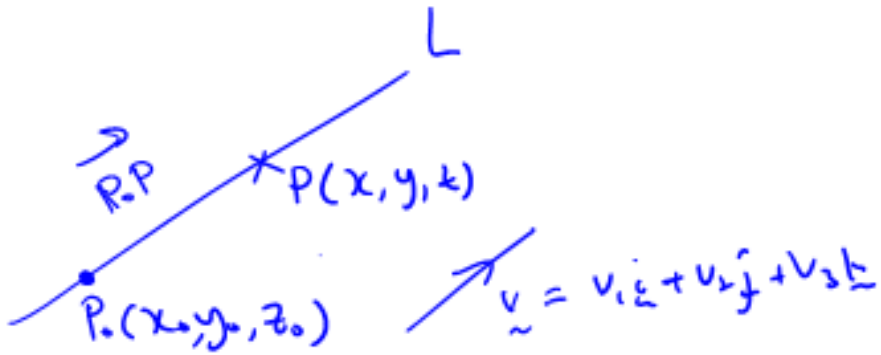
At the end of this lecture, you should be able to:

- ☐ Write the vector, parametric, and cartesian equations of a line in three-dimensional space.
- ☐ Find the distance from a point to a line in space.
- ☐ Write the vector and component equations of a plane in three-dimensional space.
- ☐ Determine the line of intersection between two planes.
- ☐ Determine the line of intersection between a line and a plane.
- ☐ Find the distance from a point to a plane.
- ☐ Calculate the angle between two planes.

Vector Equation for a Line

Suppose that L is a line in space passing through a point $P_0(x_o, y_0, z_0)$ and parallel to a vector $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$.

Let P be a point on L, then $\overrightarrow{P_0P}$ is parallel to $\boldsymbol{v}$.



$\hookrightarrow \overrightarrow{P_0P} = t\underset{\sim}{v}$ for some $t \in \mathbb{R}$.

$\overrightarrow{OP} - \overrightarrow{OP_0} = t\underset{\sim}{v}$

$\overrightarrow{OP} = \overrightarrow{OP_0} + t\underset{\sim}{v}$

$\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle v_1, v_2, v_3 \rangle$

position vector of a point on the line L.

position vector of P_0

direction vector of the line L.

Vector Equation for a Line

A vector equation for the line L through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector $\underset{\sim}{v}$ is

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + t\boldsymbol{v}, \quad -\infty < t < \infty,$$

where $\boldsymbol{r}$ is the position vector of a point $P(x, y, z)$ on L and $\boldsymbol{r}_0$ is the position vector of $P_0(x_0, y_0, z_0)$.

Parametric Equations for a Line

Parametric Equations for a Line

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$ is

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3, \quad -\infty < t < \infty.$$

vector eqn: $\langle x, y, z\rangle = \langle x_0, y_0, z_0\rangle + t\langle v_1, v_2, v_3\rangle$

$= \langle x_0 + tv_1,\ y_0 + tv_2,\ z_0 + tv_3\rangle$

Example: Find parametric equations for the line through $(-1, 0, 4)$ parallel to $\boldsymbol{v} = 2\boldsymbol{i} + 4\boldsymbol{j} - 2\boldsymbol{k}$.

v_1 v_2 v_3

x_0 y_0 z_0

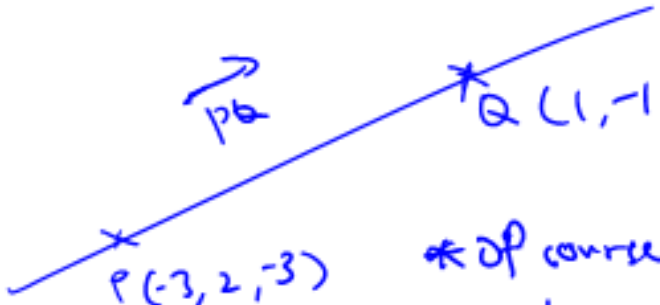
$x = -1 + 2t$

$y = 0 + 4t = 4t$, $-\infty < t < \infty$

$z = 4 - 2t$ $\quad (t \in \mathbb{R})$

Parametric Equations for a Line

Example: Find parametric equations for the line through $P(-3, 2, -3)$ and $Q(1, -1, 4)$.



The line is parallel to the vector $\vec{PQ}$

$\vec{PQ} = \vec{OQ} - \vec{OP}$

$= \langle 1, -1, 4 \rangle - \langle -3, 2, -3 \rangle$

$= \langle 4, -3, 7 \rangle$

* Of course, we can also use $\vec{QP}$ as the direction vector $\underset{\sim}{v}$.

Although they look different, both of these describe the same line.

For eg. $t_1 = 0$

$t_2 = -1$

both of them refer to the point $(-3, 2, -3)$

We can use P or Q as our P_0

Using $P(-3, 2, -3)$: $x = -3 + 4t_1$, $y = 2 - 3t_1$, $z = -3 + 7t_1$, $t_1 \in \mathbb{R}$

Using $Q(1, -1, 4)$: $x = 1 + 4t_2$, $y = -1 - 3t_2$, $z = 4 + 7t_2$, $t_2 \in \mathbb{R}$

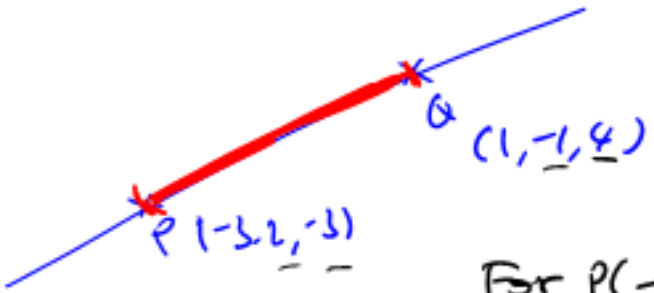
Parametric Equations for a Line Segment

If $-\infty < t < \infty$, we describe the whole straight line.

To parametrize a line segment joining two points, we first parametrize the line through the points.

Then, we find the t-value at the endpoints and restrict t to fall in the closed interval bounded by these values.

Example: Parametrize the line segment joining the points $P(-3, 2, -3)$ and $Q(1, -1, 4)$.



$x = -3 + 4t$

$y = 2 - 3t$

$z = -3 + 7t.$

$0 \le t \le 1.$

For $P(-3,2,-3)$: $-3+4t = -3 \Rightarrow t = 0$

Check : $2 - 3(0) = 2$ ✓

$-3 + 7(0) = -3.$ ✓

For $Q(1,-1,4)$: $-3+4t = 1 \Rightarrow t = 1$

Check: $2 - 3(1) = -1$ ✓ ; $-3 + 7(1) = 4$ ✓

Cartesian Equation

(Symmetrical Equation)

Cartesian Equation for a Line

The Cartesian equation of the line through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$ is

Parametric eqn: $x = x_0 + tv_1 \Rightarrow t = \frac{x-x_0}{v_1}$

$y = y_0 + tv_2 \Rightarrow t = \frac{y-y_0}{v_2}$

$z = z_0 + tv_3 \Rightarrow t = \frac{z-z_0}{v_3}$

$$\frac{x - x_0}{v_1} = \frac{y - y_0}{v_2} = \frac{z - z_0}{v_3} \quad (= t).$$

If v_1, v_2 or v_3 is 0, then x, y or z will, respectively, be a constant.

Example: Find the Cartesian equation for the line through $(2, -3, 1)$ parallel to $\boldsymbol{v} = 3\boldsymbol{i} + 6\boldsymbol{k}$.

Parametric eqn: $x = 2 + 3t \Rightarrow t = \frac{x-2}{3}$

$y = -3 + 0t \Rightarrow y = -3$, $t \in \mathbb{R}$.

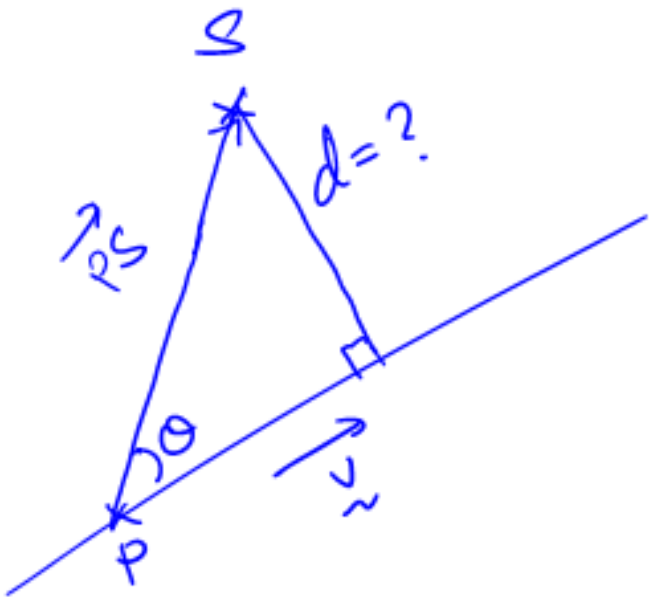
$z = 1 + 6t \Rightarrow t = \frac{z-1}{6}$

Cartesian eqn: $\frac{x-2}{3} = \frac{z-1}{6}$, $y = -3$

Distance from a Point to a Line

(Shortest)

Distance from a point S to a Line Through P Parallel to $\boldsymbol{v}$



$$d = \frac{|\overrightarrow{PS} \times \boldsymbol{v}|}{|\boldsymbol{v}|}$$

Let θ be the angle between $\vec{PS}$ and $\underset{\sim}{v}$

$$d = |\vec{PS}| \sin\theta.$$

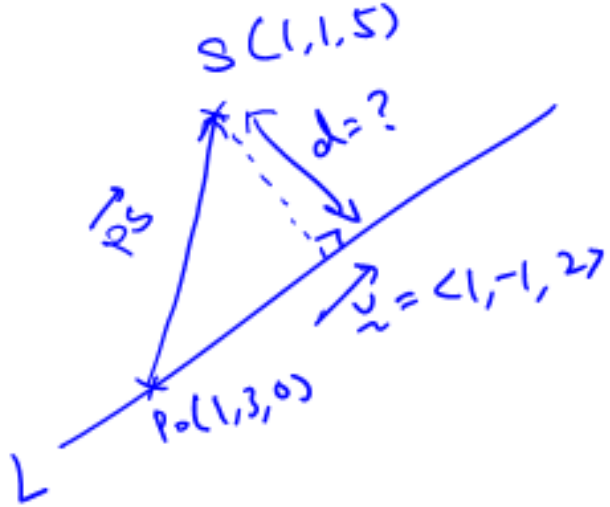
$$= \frac{|\vec{PS}||\underset{\sim}{v}| \sin\theta}{|\underset{\sim}{v}|}$$

$$= \frac{|\vec{PS} \times \underset{\sim}{v}|}{|\underset{\sim}{v}|}$$

Distance from a Point to a Line

Example: Find the distance from the point $S(1,1,5)$ to the line

$$L: \quad x = 1 + t, \quad y = 3 - t, \quad z = 2t. \qquad = 0 + 2t$$



$\hookrightarrow P_0(1,3,0)$

$\underset{\sim}{v} = \langle 1,-1,2 \rangle$

$\overrightarrow{P_0S} = \langle 1,1,5 \rangle - \langle 1,3,0 \rangle$

$= \langle 0,-2,5 \rangle$

$$\overrightarrow{P_0S} \times \underset{\sim}{v} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = \begin{vmatrix} -2 & 5 \\ -1 & 2 \end{vmatrix} \underset{\sim}{i} - \begin{vmatrix} 0 & 5 \\ 1 & 2 \end{vmatrix} \underset{\sim}{j} + \begin{vmatrix} 0 & -2 \\ 1 & -1 \end{vmatrix} \underset{\sim}{k}$$

$= \underset{\sim}{i} + 5\underset{\sim}{j} + 2\underset{\sim}{k}$

The distance, $d = \dfrac{\left|\overrightarrow{P_0S} \times \underset{\sim}{v}\right|}{|\underset{\sim}{v}|}$

$= \dfrac{\sqrt{(1)^2 + 5^2 + 2^2}}{\sqrt{1^2 + (-1)^2 + 2^2}} = \sqrt{5}$ #

Equation for a Plane in Space

Equation for a Plane

The plane through $P_0(x_0, y_0, z_0)$ normal to a nonzero vector $\boldsymbol{n} = A\boldsymbol{i} + B\boldsymbol{j} + C\boldsymbol{k}$ has

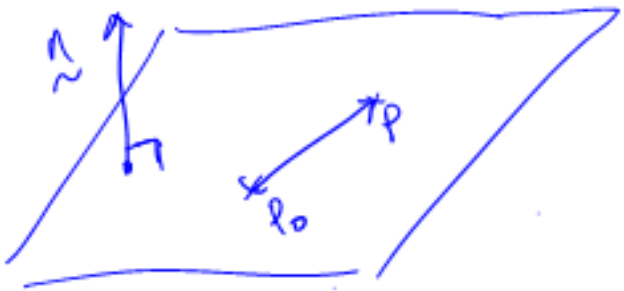
$\underset{\sim}{n} = \langle A, B, C \rangle$

$$\text{Vector equation : } \boldsymbol{n} \cdot \overrightarrow{P_0P} = 0$$

$$\text{Component equation : } A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

$$\text{Component equation simplified : } Ax + By + Cz = D, \text{ where}$$

$$D = Ax_0 + By_0 + Cz_0$$



Let $P(x, y, z)$ be any point on the plane.

$\Rightarrow \overrightarrow{P_0P}$ is a vector parallel to the plane.

$\Rightarrow \underset{\sim}{n}$ is orthogonal to $\overrightarrow{P_0P}$

$\Rightarrow \underset{\sim}{n} \cdot \overrightarrow{P_0P} = 0$

$\Rightarrow \langle A, B, C \rangle \cdot (\overrightarrow{OP} - \overrightarrow{OP_0}) = 0$

$\Rightarrow \langle A, B, C \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$

$$\Rightarrow A(x-x_0) + B(y-y_0) + C(z-z_0) = 0$$

$$\Rightarrow Ax - Ax_0 + By - By_0 + Cz - Cz_0 = 0$$

$$\Rightarrow Ax + By + Cz = Ax_0 + By_0 + Cz_0 = D$$

Equation for a Plane in Space

Example: Find an equation for the plane through $P_0(-3, 0, 7)$ perpendicular to $\boldsymbol{n} = 5\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$.

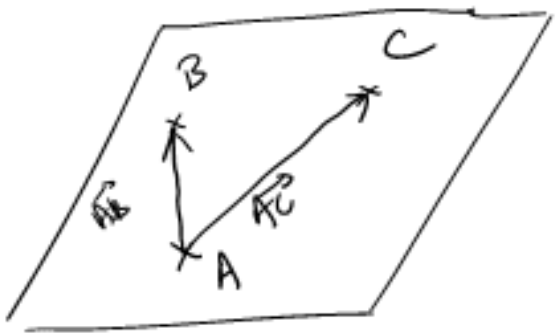
Component Equation: $5(x-(-3)) + 2(y-0) + (-1)(z-7) = 0$

$$5x + 15 + 2y - z + 7 = 0$$

$$5x + 2y - z = -22.$$

Equation for a Plane in Space

Example: Find an equation for the plane through $A(0,0,1)$, $B(2,0,0)$, and $C(0,3,0)$.



A vector perpendicular to the plane span by $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is

$$\underset{\sim}{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \langle 2,0,-1 \rangle \times \langle 0,3,-1 \rangle$$

$$= \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -1 \\ 3 & -1 \end{vmatrix} \underset{\sim}{i} - \begin{vmatrix} 2 & -1 \\ 0 & -1 \end{vmatrix} \underset{\sim}{j} + \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} \underset{\sim}{k}$$

$$= 3\underset{\sim}{i} + 2\underset{\sim}{j} + 6\underset{\sim}{k}$$

Equation of the plane:

$$3(x-0) + 2(y-0) + 6(z-1) = 0$$

$$3x + 2y + 6z = 6$$

Alternatively, you can also use

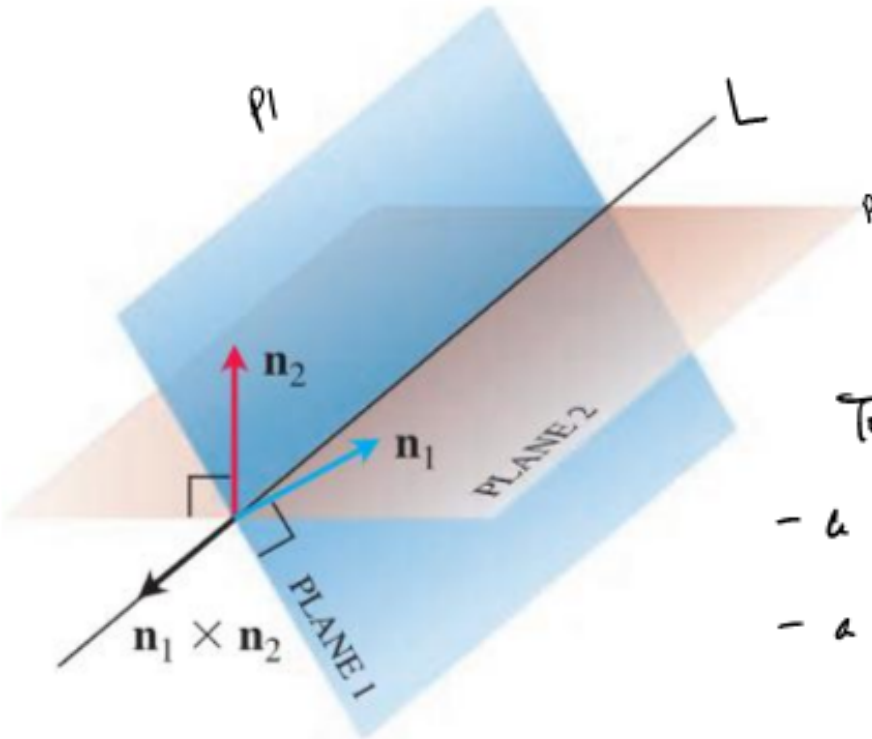
$\underset{\sim}{n} = \overrightarrow{BA} \times \overrightarrow{BC}$ or $\underset{\sim}{n} = \overrightarrow{CA} \times \overrightarrow{CB}$

Intersection of Two Planes

(Their direction vector are parallel, i.e. $\underset{\sim}{v_1} = k\underset{\sim}{v_2}$ for some scalar k)

Just as lines are parallel if and only if they have the same direction, two planes are parallel if and only if their normals are parallel.

Two planes that are not parallel intersect in a line.



Let L be the line of intersection of the plane P_1 and the plane P_2.

L must lie on both P_1 and P_2.

$\Rightarrow$ L is orthogonal to $\underset{\sim}{n_1}$ and $\underset{\sim}{n_2}$

$\Rightarrow$ L is parallel to $\underset{\sim}{n_1} \times \underset{\sim}{n_2}$

To find the equation of the line L, we need

- a point on L (Take any point lies on both planes)
- a direction vector $(\underset{\sim}{n_1} \times \underset{\sim}{n_2})$

Intersection of Two Planes

Example: Find the equation of the line of intersection of the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

$\underset{\sim}{n_1} = \langle 3, -6, -2 \rangle$ $\quad$ $\underset{\sim}{n_2} = \langle 2, 1, -2 \rangle$

$$\underset{\sim}{n_1} \times \underset{\sim}{n_2} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = \begin{vmatrix} -6 & -2 \\ 1 & -2 \end{vmatrix} \underset{\sim}{i} - \begin{vmatrix} 3 & -2 \\ 2 & -2 \end{vmatrix} \underset{\sim}{j} + \begin{vmatrix} 3 & -6 \\ 2 & 1 \end{vmatrix} \underset{\sim}{k}$$

$$= 14\underset{\sim}{i} + 2\underset{\sim}{j} + 15\underset{\sim}{k}$$ ← the direction vector of the line of intersection.

Let $P(a, b, c)$ be a point on the line of intersection.

So, $3a - 6b - 2c = 15$

$2a + b - 2c = 5$

Let $a = 0$, $-6b - 2c = 15$ — ①

$b - 2c = 5$ — ②

② − ①, $7b = -10 \Rightarrow b = \frac{-10}{7}$

From ②, $\frac{-10}{7} - 2c = 5 \Rightarrow c = \frac{-45}{14}$

Of course you may also let $b = 0$ or $c = 0$ or any numbers.

So, $P(0, \frac{-10}{7}, \frac{-45}{14})$ is a point on the line of intersection. So the equation of the line of intersection: $x = 0 + 14t = 14t$, $y = \frac{-10}{7} + 2t$, $z = \frac{-45}{14} + 15t$, $t \in \mathbb{R}$.

Intersection of a Line and a Plane

Example: Find the point where line

$$x = \frac{8}{3} + 2t, \quad y = -2t, \quad z = 1 + t$$

intersects the plane $3x + 2y + 6z = 6$.

Substitute the eqn of line into eqn. of plane.

$$3\left(\tfrac{8}{3} + 2t\right) + 2(-2t) + 6(1+t) = 6$$

$$8 + 6t - 4t + 6 + 6t = 6$$

$$t = -1$$

← the intersection happens at $t = -1$.

So the point of intersection is

$$\left(\tfrac{8}{3} + 2(-1),\ -2(-1),\ 1+(-1)\right) = \left(\tfrac{2}{3}, 2, 0\right)$$

Intersection of a Line and a Plane

Example: Determine where the line

$$x = -1 + t, \quad y = 2 + 5t, \quad z = 5 + t$$

intersects the plane $x - y + 4z = 5$.

$$(-1+t) - (2+5t) + 4(5+t) = 5$$

$$-1+t-2-5t+20+4t = 5$$

$$17 = 5 \quad \text{(contradiction)}$$

the intersection does not happen for any t.

The line does not intersect with the plane.

Intersection of a Line and a Plane

Example: Determine where the line

$$x = -1 + t, \quad y = 2 + 5t, \quad z = 5 + t$$

intersects the plane $x - y + 4z = 17$.

$(-1+t) - (2+5t) + 4(5+t) = 17$

$\vdots$

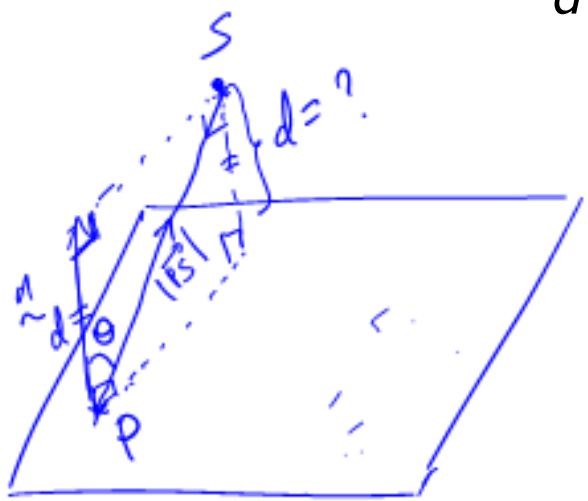
$17 = 17$ (always valid) intersection happens at all t

So, the line and the plane intersect at the line itself.
(the line lies on the plane).

Distance from a Point to a Plane

(Shortest)

Distance from a Point S to a Plane Through a Point P with a Normal $\underset{\sim}{n}$



$$d = \left| \overrightarrow{PS} \cdot \frac{\underset{\sim}{n}}{|\underset{\sim}{n}|} \right| \qquad \frac{|\overrightarrow{PS} \cdot \underset{\sim}{n}|}{|\underset{\sim}{n}|}$$

Let θ = acute angle between $\overrightarrow{PS}$ and $\underset{\sim}{n}$.

$$\cos\theta = \frac{d}{|\overrightarrow{PS}|}$$

$$d = |\overrightarrow{PS}| \cos\theta$$

$$= \frac{|\overrightarrow{PS}|\,|\underset{\sim}{n}| \cos\theta}{|\underset{\sim}{n}|}$$

$$= \frac{|\overrightarrow{PS} \cdot \underset{\sim}{n}|}{|\underset{\sim}{n}|}$$

$|\overrightarrow{PS} \cdot \underset{\sim}{n}| = |\overrightarrow{PS}|\,|\underset{\sim}{n}| \cos\theta$.

$0 \le \theta \le \pi/2$.

Distance from a Point to a Plane

Example: Find the distance from $S(1,1,3)$ to the plane $3x+2y+6z=6$.

$\underset{\sim}{n} = \langle 3, 2, 6 \rangle$

First, we need to find a point on the plane.

Let $x=0$, $y=0 \Rightarrow 3(0)+2(0)+6z=6$

$z=1$

So $P(0,0,1)$ is a point on the plane.

$\overrightarrow{PS} = \langle 1, 1, 2 \rangle$

$$d = \frac{|\overrightarrow{PS} \cdot \underset{\sim}{n}|}{|\underset{\sim}{n}|} = \frac{|(1)(3)+(1)(2)+(2)(6)|}{\sqrt{3^2+2^2+6^2}} = \frac{17}{7}.$$

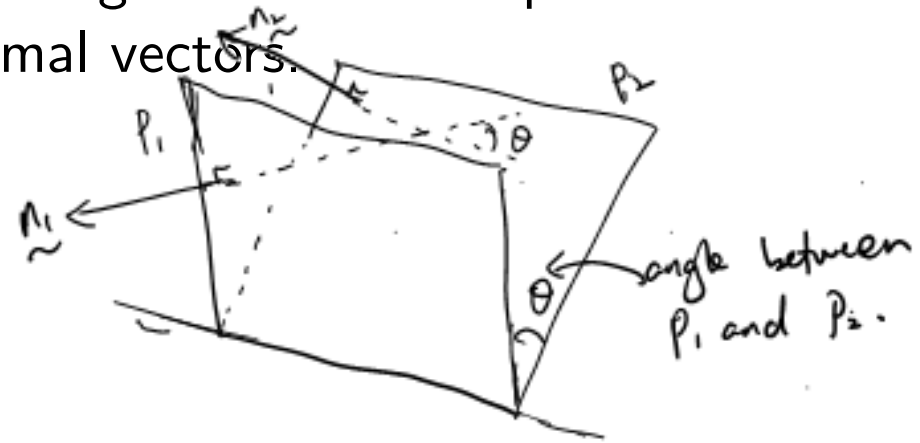
Distance from a Point to a Plane

Example: Find the distance from $S(2, -1, 3)$ to the plane $3x - 26y + 7z = -90$.

Exercise : (Ans : $\frac{143}{\sqrt{734}}$)

Angles Between Planes

The angle between two planes is the same as the angle between their normal vectors.



$$\cos\theta = \frac{n_1 \cdot n_2}{|n_1||n_2|}$$

$0 \le \theta \le \pi$.

Example: Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

$n_1 = \langle 3, -6, -2 \rangle$

$n_2 = \langle 2, 1, -2 \rangle$

$$\cos\theta = \frac{\langle 3, -6, -2 \rangle \cdot \langle 2, 1, -2 \rangle}{|n_1||n_2|}$$

$$= \frac{(3)(2) + (-6)(1) + (-2)(-2)}{\left(\sqrt{3^2 + (-6)^2 + (-2)^2}\right)\left(\sqrt{(2)^2 + 1^2 + (-2)^2}\right)} = \frac{4}{21}$$

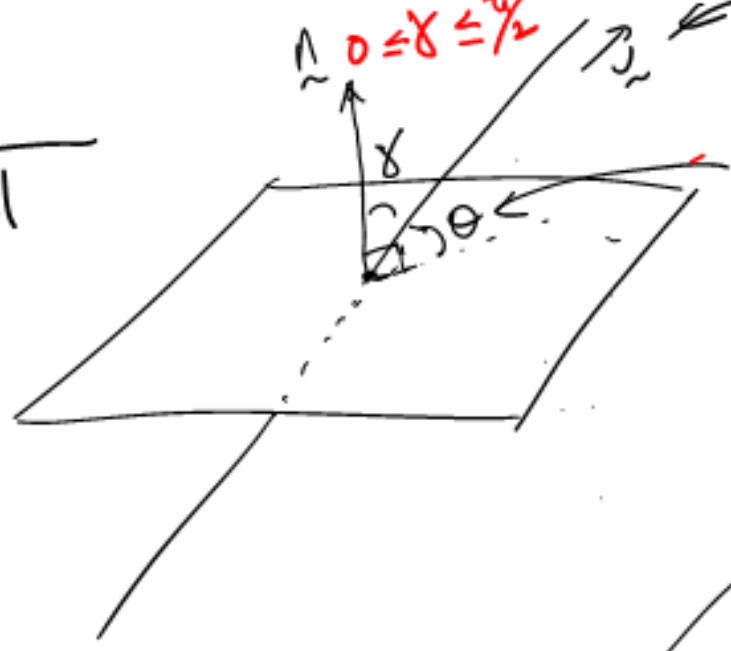
$\theta = \arccos\left(\frac{4}{21}\right) \approx 1.38$ rad.

Angle between line & plane

$\underset{\sim}{v}$ = direction vector of the line.

Case 1:

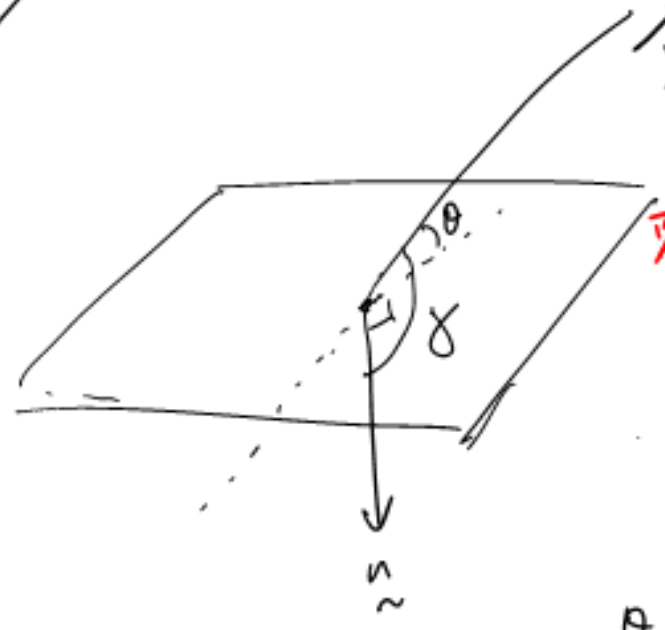
$$\cos\gamma = \frac{\underset{\sim}{n}\cdot\underset{\sim}{v}}{|\underset{\sim}{n}||\underset{\sim}{v}|}$$



angle between line and plane.

$$\theta = \frac{\pi}{2} - \gamma$$

Case 2:



$\pi/2 < \gamma \le \pi$

$$\cos\gamma = \frac{\underset{\sim}{n}\cdot\underset{\sim}{v}}{|\underset{\sim}{n}||\underset{\sim}{v}|}$$

$$\theta = \gamma - \pi/2.$$